$d(T) = \frac{1}{1+rT}$ (simple) $= \frac{1}{(1+\frac{r[m]}{m})^{mT}}$ (SAC) $= e^{-r[\infty]\cdot T}$ (continuous) $= \frac{1}{(1+R)^T}$ (effective.)

$f(t) = r_{t-\frac{1}{2},t}$, $\left(1+\frac{\hat f(T)}{2}\right)^{2T} = \left(1+\frac{f(1/2)}{2}\right)\left(1+\frac{f(1)}{2}\right)\cdots\left(1+\frac{f(T)}{2}\right)$ $\hat f(\frac{N}{2}) \approx \frac{1}{N}\sum_{i=1}^{N} f(i/2)$

borrow/receive, buy ZCB, $t=0$, $P = A\,d(\eta) - F\,d(T) = 0$; deposit of \$1 at $t-\frac{1}{2}$, get $\left(1+\frac{f(t)}{2}\right)$ at t

sell ZCB pay, $\Rightarrow F = A\frac{d(\eta)}{d(T)} = A\left(1+\frac{r_{0,\eta,T}}{2}\right)^{2(T-\eta)}$ (for $r_{0,\eta,T}$) ; $\Rightarrow f(t) = 2\left(\frac{d(t-1/2)}{d(t)}-1\right)$

$P^{CB} = \frac{qF}{2}\sum_{i=1}^{2T}\lambda^i + F\lambda^{2T}$, $\lambda = \frac{1}{1+y/2}$, $\frac{2}{y} = \frac{\lambda}{1-\lambda}$, $\sum_{i=1}^{2T}\lambda^i = \lambda\left(\frac{1-\lambda^{2T}}{1-\lambda}\right)$ YTM y depends only on coupon rate, not A or F.

[CB] = [A] + [ZCB], $\min(y_A, y_{ZCB}) \le y_{CB} \le \max(y_A, y_{ZCB})$, $\min(\hat f(i)) < y_A < \max(\hat f(i))$

→ if $f(T+1/2) > q$, keeping F, q the same & increasing $T \to T+1/2$ will decrease P.

→ bond trades at par if $q=y$, above par if $q>y$.

if spot rate $\hat f(t)$ upward sloping, $y_A \le y_{PC} \le \hat f(t)$ & y_A, y_{PC} also upward sloping. $\hat f(t) \ge \hat f(t-1/2) \Leftrightarrow f(t) \ge \hat f(t)$ $\Leftrightarrow f(t) \ge \hat f(t-1/2)$ (incr. can be strict)

price of PAR COUPON $= F = F\,d(T) + F\frac{q_{PC}(T)}{2}\sum_{i=1}^{2T} d(\frac{i}{2})$ (& $f(t)\nearrow \Rightarrow \hat f(t)\nearrow$) & $q\uparrow \Rightarrow y\downarrow$

perpetuity (annuity w/o maturity): $y_P = 2A/P$. $\Delta P = f'(y)\Delta y + \frac{1}{2}f''(y)(\Delta y)^2$, $\Delta P^{ZCB} = -\frac{PT}{(1+y/2)}\Delta y$ (% → bp)

[ZCB] $DV01 = \frac{1}{10k}\left(\frac{FT}{(1+y/2)^{2T+1}}\right)$, $D = \frac{T}{1+y/2}$, $D_{MAC} = T$, $C = \frac{T(T+1/2)}{(1+y/2)^2}$, $T^* = \frac{1}{2\ln(1+y/2)} \approx \frac{1}{y}$

[CB] $DV01 = \frac{F}{10k}\left(\frac{q}{y^2}\left(1-\frac{1}{(1+y/2)^{2T}}\right) - \left(1-\frac{q}{y}\right)\frac{T}{(1+y/2)^{2T+1}}\right)$, $D_{MAC} = \frac{F}{P}\left(\frac{T}{(1+y/2)^{2T}} + \frac{q}{2}\sum_{i=1}^{2T}\frac{i/2}{(1+y/2)^i}\right)$ DV01 = dollar value of 1bp

$D_{MAC} = \frac{F}{P}(1+y/2)\left(\frac{q}{y^2}\left(1-\frac{1}{(1+y/2)^{2T}}\right) + \left(1-\frac{q}{y}\right)\frac{T}{(1+y/2)^{2T+1}}\right)$ $DV01^{CB} > DV01^{ZCB}$, $D^{CB} < D^{ZCB}$ (C too) avg. of T is earlier (wt) more pymts @ time < T

let $\lambda = \frac{1}{1+y/2}$, $C = \frac{F\lambda^2}{P}\left(T(T+1/2)\lambda^{2T} + \frac{q}{2}\sum_{i=1}^{2T}\frac{i(i+1)\lambda^i}{4}\right) = \frac{F}{P}\left(\frac{2q}{y^3}(1-\lambda^{2T}) - \frac{2q}{y^2}T\lambda^{2T+1} + \left(1-\frac{q}{y}\right)T(T+1/2)\right)$ $\cdot\lambda^{2T+2}$

→ sps $q^2 > q^1$, $B^2 = B^1 + [A] \Rightarrow DV01^2 > DV01^1$, $D^1 > D^A$ (↑weight on final pymt) $\Rightarrow D^2 < D^1$

→ if $q \ge y$, D↑ with T, if $q<y$, D↑ until T^*, then decreases. (DV01 too). (C too) incr. q: ↑DV01, ↓D, ↓C, incr. y: ↓DV01, D, C

[A] $D_{MAC} = \frac{1}{y} + \frac{1}{2} - \frac{T}{(1+y/2)^{2T}-1}$, $DV01 = \frac{1}{10k}\frac{2A}{y}(1-\lambda^{2T})\left(\frac{1}{y} - \frac{\lambda T}{(1+y/2)^{2T}-1}\right)$, note $P = \frac{2A}{y}(1-\lambda^{2T})$

[par] $DV01 = \frac{F}{(10k)}y(1-\lambda^{2T})$, $D_{MAC} = \left(\frac{1}{y}+\frac{1}{2}\right)(1-\lambda^{2T})$, [perpetuity] $DV01 = \frac{2A}{(10k)y^2}$, $D_{MAC} = \frac{1}{2}+\frac{1}{y}$, $C = \frac{2}{y^2}$

$D = -\frac{1}{P}\frac{dP}{dy}$, $DV01 = \frac{PD}{10{,}000}$, $C = +\frac{1}{P}\frac{d^2P}{dy^2}$, $D = \frac{p^1D^1 + p^2D^2}{p^1+p^2}$ ← weighted avg (sim. to C), $DV01 = DV01^1 + DV01^2$ ← add.

$P^{float} = F$ just after pymnt (& behaves like ZCB wrt D and C). ↓1/2 $C>0 \Leftrightarrow$ price higher than linear approx.

Swap: receiver = side of contract receiving fixed pymt, plain vanilla replicated by long par coupon & short float.

interest FRB: face F at maturity and periodic pymts $\frac{F}{2}(q^* - r_{t-\frac{1}{2},t})$

↳ let $q^* = 2q$, then [iFRB] + [float] = 2[par CB] $\Rightarrow P^{iFRB} = F$, $2\cdot F\cdot D^{par} = \cancel{F\cdot D^{float}}$ (~0) $+ F\cdot D^{iFRB} \Rightarrow D^{iFRB} \approx 2D^{par}$

accrued interest = [principle] × [rate] × [time] = $\frac{qF}{2}\cdot\frac{\text{days elapsed since last pymt}}{\text{days between pymts}}$, $P^{full} = P^{flat} + AI$

↳ $P^{full} = F\lambda^{k+\tau} + \frac{qF}{2}\sum_{i=0}^{k}\lambda^{i+\tau} = \lambda^\tau\left(\frac{qF}{2} + F\lambda^k + \frac{qF}{2}\sum_{i=1}^{k}\lambda^i\right)$, k = # pymts remaining aft. next, ↑ quoted price

callable bond: CB + issuer has option to buy back for F, τ = fraction of current coupon remaining → call option

↳ P^c, P^B = price of callable/non-callable bond, at time T_E, $P^c = \min(P^B, F) = P^B - \max(P^B - F, 0)$

neg. convexity/price compression: convex yield curve, price gain slows as yield fall

↳ approach call price.

- bullet (single maturity) vs. barbell (2 diff. T, higher C, long volatility) → make profit if volatility goes up.
  ↳ butterfly state: αP, $-P$, $(1-\alpha)P$ in T^1, T^2, T^3, choose α s.t. $D^{bull} = D^{bar}$
- first order estimate: $\Delta P \approx -DV01\,\Delta y$, 2nd order: $\Delta P \approx -PD\,\Delta y + \frac{1}{2}PC(\Delta y)^2$

Ex: A pay B $\frac{1}{4}F\cdot r_{t-\frac{1}{2},t}$ for $t=.5$ to 5, $\frac{3}{4}F\cdot r_{t-\frac{1}{2},t}$ for $t=5$ to 10, B pays A $\frac{qF}{2}$. no pymt at $t=0$ or T.

replicate B's position. [long float] hold float ≡ face $\frac{F}{2}$ from [0,5], $\frac{3F}{2}$ from [5,10]. face $3F/2$, maturity 10

↳ hold $F/2$ at $t=0$, short ZCB ≡ face $F/2$, maturity 5. long ZCB ≡ face $3F/2$, maturity 5, short ZCB ↑

[fixed] short CB ≡ coupon rate q, maturity 10 & long ZCB, face F maturity 10.

then at $t=0$, $\left(\frac{F}{2} - \frac{F}{2}d(5)\right) + \left(\frac{3F}{2}d(5) - \frac{3F}{2}d(10)\right) - \left(Fd(10) + \frac{qF}{2}\sum_{i=1}^{20}d(i/2)\right) + Fd(10) = 0$.

Ex: $\hat f(t)$ incr, (1) par, T=10, (2) CB below par, T=10, (3) A, T=7 (4) ZCB, T=7, (5) ZCB, T=10.

[YTM] $y_a^{10} < \hat f(10) = y^5$, similarly $y_3 < y_4$, [2] = [A¹⁰] + [5] $\Rightarrow y_a^{10} < y_2 < y_5$, $y_a^{10} < y_1 < y_2$ & $y_4 < y_5$

[A¹⁰] = [A⁷] + [eq. pymts from 7 to 10], y s.t. $\hat f(7.5) < y < \hat f(10) \Rightarrow y_3 < y$, $y_3 < y_a^{10} < y_1 < y_2 < y_5$ & $y_3 < y_4 < y_5$

[convexity] under typical market conditions, C inc. with maturity (regardless of y), since C is a weighted avg ↙ same avg applies

[D_{MAC}] (1) CB ≡ $q=1.2y$, T=10, (2) par, T=10, (3) A^{10}, pymt $\frac{yF}{2}$, (4) A, pymt $\frac{yF}{2}$ at $t=.5, 1.5, \ldots, 9.5$

$D^3 < 10$, [1] = [3] + [ZCB¹⁰] → $D^3 < D^1 < 10$, $D^1 < D^2 < 10$; [3] = [4] + [A pymts at t=1,2,...,10] → D^5 D is weighted avg.

$D^5 - D^4 = \sum_{i=1}^{10}\frac{A\lambda^{2i}\cdot i}{A\sum_{j=1}^{10}\lambda^{2j}} - \sum_{i=1}^{10}\frac{A\lambda^{2i-1}\cdot\frac{1}{2}}{A\sum_{j=1}^{10}\lambda^{2j-1}} = \sum_{i=1}^{10}\frac{i/2}{(1+\frac{1}{2})^{2i}\sum_{j=1}^{10}\lambda^{2j}} > 0$, then $D^4 < D^3 < D^5$ (& DV01 $4<3<2<1$) (<0, >0)

if $r_{new}(5) > r_{new}(2), r_{new}(10)$, short 5 (long 2,10), $\Delta P = -F_2^5 DV01^5\Delta r^5 - F^2 DV01^2\Delta r^2 - \ldots$ (>0) be ↑pymt ↑DV01 $(\Delta r^5 - \Delta r^2)$

[float] each pymt = forward loan w/o face pymt, long ZCB^F, maturity $\frac{i-1}{2}$, short $F/\frac{i}{2}$, for. at rate $f(i/2)$

↳ value of C_i at $t=0$ is $F\left(d(\frac{i-1}{2}) - d(\frac{i}{2})\right) + 0 \Rightarrow Fd(T) + \sum_{i=1}^{2T}(\ldots) = F$ $d(0)=1$, $\hat f(1/2) = f(1/2)$.

Ex: [T-Bills] if $n \le 182$, $(100-P) = \underbrace{P}_{\text{principal}}\cdot\underbrace{y_{be}}_{\text{interest}}\cdot\underbrace{\frac{n}{365}}_{\text{time}}$, for $n>182$, $P^{full} = 100$, $P^{flat} = P\left(1+\frac{y_{bk}}{2}\right)$ (SAC), $AI = P^{flat}\cdot y_{be}\cdot\frac{n-\frac{365}{2}}{365}$

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yield curves: [reference yield] + [perturbation] → [new yield curve] → [new discount factor] → [new P]

↳ new yield $= y(t) + (y_i - y_i^*)Y_i(t)$, y_i^* the original value, y_i the new value of ith key rate (time after 1/2 yr ↗)

then if security has $P = f(x_1, \ldots, x_4)$, matching key rate DV01 $\Leftrightarrow \frac{\partial f}{\partial x_i}(\vec 0) = \frac{\partial g}{\partial x_i}(\vec 0)$ (in yield)

PCA: new yield curve $y(t) + Z_i Z(t) \Rightarrow Z_i(t)$ statistically most significant based on observed changes

(1) parallel shift (direction in yield curve space ≡ most variance), (2) slope, (3) curvature shift.

time $n \in [0,m]$ price of pymt W_m @ $t=m$, $W_n = \tilde E_n\left[\frac{D_n}{D_m}W_m\right]$, $D_n = \frac{1}{(1+R_0)\cdots(1+R_{n-1})}$, $D_0 = 1$.

↳ $\tilde E_n[X] = \tilde E_n[\tilde E_n[X]]$ if $n \le m \le N$, $E_n[\phi(X)] \ge \phi(\tilde E_n[X])$ for ϕ convex. $\phi(\lambda x) \le \lambda\phi(x)$

[ZCB] $B_{n,m} = \frac{1}{D_n}\tilde E_n[D_m]$ = price of \$1 face ZCB (maturity m) at time n

↳ if security pays C_k at time k, until m, $P_0 = \tilde E\left[\sum_{k=1}^{m}D_kC_k\right] = \sum_{k=1}^{m}C_kB_{0,k}$ $A_{n,m} = \sum_{k=n+1}^{m}B_{n,k}$, $C_{n,m} = q\sum_{k=n+1}^{m}B_{n,k} + B_{n,m}$ after pymt at time n

time-0 value of R_{n-1} paid at $n = \tilde E[D_nR_{n-1}] = B_{0,n-1} - B_{0,n}$ = loan \$1, receive interest, repay @ n.

- n-period interest rate swap ≡ fixed pymt K = pymts $K - R_{n-1}$ @ $n = 1,2,\ldots,m$ caplets/floorlets make 1 pymt.
- cap/floor = $\ldots (R_{n-1}-K)^+ / (K-R_{n-1})^+$ ↔ floor = swap + cap
- swap$^K_{0,m} = \sum_{i=1}^{m}\tilde E[D_i(K-R_{i-1})] = C^K_{0,m} - 1 = K\sum_{i=1}^{m}B_{0,i} + B_{0,m} - 1$, swap rate = K s.t. swap$^K_{0,m} = 0$.

backward induction: time-n value after pymt C_n, $V_n = \frac{1}{1+R_n}\tilde E_n[C_{n+1} + V_{n+1}]$. w/o pymt, D_nV_n is martingale ↳ $V_n = \frac{\tilde E_n[V_{n+1}]}{1+R_n}$

→ American option: expiration m, adapted process $(G_n)_{0\le n\le m}$ (intrinsic value process), sps $G_n \ge 0$

- at each $n=0,\dots,m$, choose whether to exercise G_n, only 1 chance to exercise.
- at m, can no longer wait, $V_m = G_m$, otherwise $V_n = \max\{G_n, \frac{1}{1+R_n}\tilde{E}_n[V_{n+1}]\}$. s.t. $V_n = G_n$.

Bermudan option: $\phi \ne \mathcal{E} \subset \{0,\dots,N\}$ of possible exercise dates ↳ alw. optimal to exercise at smallest n

→ let $(F_n)_{n\in\mathcal{E}}$ be a set of possible call/put prices, if option of puttable/callable bond exercised, holder of bond receives $Fq + F_n$ & no further pymts: issuer (holder) of bond has right for call/put. (-able bond).

$V_n^c = \min\{F_n, \frac{1}{1+R_n}\tilde{E}_n[V_{n+1}^c + qF]\}$, max for put, second term only if $n \notin \mathcal{E}$.

* let U, W be Bermudan options, $G_n^U = (FC_{n,m}^q - F_n)^+$, $G_n^W = (F_n - FC_{n,m}^q)^+$, $V_n^c = FC_{n,m}^q - U_n$, $V_n^p = FC_{n,m}^q + W_n$
* let $P_n \ge 0$ be an adapted process s.t. D_nP_n is a martingale, $G_n = (P_n - K)^+$

then it is never beneficial to exercise V before m ↔ if $n \le m-1$, $\frac{1}{1+R_n}\tilde{E}_n[V_{n+1}] \ge (P_n - K)^+$ ⇕

→ $D_nP_n = \tilde{E}_n[D_{n+1}P_{n+1}]$, $\phi(0)=0$ (works for call but not put.), ϕ convex non-neg, $G_n = \phi(P_n)$, V American, then never optimal ... before m

→ Forward contract: short position agrees to sell asset at time m at delivery price K, long ... buy

- $For_{n,m}$ = value of K s.t. price of both positions zero at time n, $For_{0,m} = K$. (actual K).
- $For_{n,m} = \frac{\tilde{E}_n[D_mP_m]}{\tilde{E}_n[D_m]}$, if no dividends/coupons, $\frac{D_nP_n}{D_nB_{n,m}} = \frac{P_n}{B_{n,m}}$ ↔ $\frac{1}{P_n}\tilde{E}_n[D_m(P_m - K)] = 0$. value of long at maturity is $P_m - K$. $(D_nB_{n,m}For_{n,m})$ is a martingale.

→ Futures: all contracts maturing at m have the same delivery price → $(Fut_{n,m})_{0\le n\le m}$, s.t.

(1) $Fut_{m,m} = P_m$ (2) $\tilde{E}_n[\sum_{i=n}^{m-1} D_{i+1}(Fut_{i+1,m} - Fut_{i,m})] = 0$ for $n=0,\dots,m-1$. → at each n, value of future stream of adj. is zero.

long ↔ receive $Fut_{k+1,m} - Fut_{k,m}$ on day $(k+1)$ for $k=0,\dots,m-1$.

* $Fut_{n,m} = \tilde{E}_n[P_m]$ is a martingale, and $\sum_{i=0}^{m-1}(Fut_{i+1,m} - Fut_{i,m}) = P_m - Fut_{0,m}$

→ let $F_{n,m}$ = forward rate agreed to at n, borrowing from m to $m+1$, $For_{n,m} = \frac{\tilde{E}_n[D_mB_{m,m+1}]}{\tilde{E}_n[D_m]} = \frac{1}{1+F_{n,m}}$

bc (1) $\tilde{E}_n[D_mB_{m,k}] = D_nB_{n,k}$, (2) $\tilde{E}_n[D_m] = \tilde{E}_n[D_mB_{m,m}] = D_nB_{n,m}$ ↑ 1-period ZCB.

also, $Fut_{n,m} = \tilde{E}_n[B_{m,m+1}] = \tilde{E}_n[\frac{1}{1+R_m}] \xrightarrow{\text{Jensen}} Fut_{n,m} > \frac{1}{1+\tilde{E}_n[R_m]}$ ↗ $R_{n,m}^{fut} = \tilde{E}_n[R_m] > F_{n,m}$

X, Y positively correlated if $\tilde{E}[XY] > \tilde{E}[X]\tilde{E}[Y]$, if D_m & P_m positively correlated, $For_{n,m} > Fut_{n,m}$

→ Interest rate future: Eurodollar $QP_{n,m} = 100(1 - kR_m)$, $k=4$ (3 mth LIBOR).

→ mortage rate y = YTM, $\lambda = \frac{1}{1+y/12}$, then $P = A\sum_{i=1}^{12T}\lambda^i = \frac{A\lambda(1-\lambda^{12T})}{1-\lambda}$, A = monthly pymt.

sps constant monthly repayment rate SMM, then annual rate $CPR = 1-(1-SMM)^{12}$

prepayment model: $r_n = \alpha + \beta(Y - \mu - R_n)^+$, → due to refinancing ↳ idea: when rates high, no incentive to refinance.

$I_t = rB_{t-1}$ → scheduled $P_t^s = A_t - I_t$ → pre-pymt $P_t^p = r_t(B_{t-1} - P_t^s)$ → $A_{t+1} = A_t(1-r_t)$

note the remaining pymts $B_t = B_{t-1} - P_t^s - P_t^p$ ($A_0 = A$, $B_0 = P$).

↳ passthroughs: $C_i = I_i + P_i^s + P_i^p$, IO/PO strips (derivatives) only receive $I_i / P_i^s + P_i^p$.